Zixiao Jin

School of Computer Science, China University of Geosciences (Wuhan), China
jinzixiao123@cug.edu.cn | Homepage | Google Scholar

RESEARCH INTERESTS

1. Multimodal representation learning, clustering, and disentanglement.
2. Graph and hypergraph learning for biomedical data, including single-cell multi-omics and spatial transcriptomics.
3. Explainability and reasoning in multimodal large language models (MLLMs) and vision–language models (VLMs).

EDUCATION

China University of Geosciences (Wuhan) Wuhan, China
M.S. in Computer Science and Technology; Advisor: ***Prof. Chang Tang***
Sep. 2024 – Expected Jun. 2027 *Average Score (100-point scale): 87.72/100*

China University of Geosciences (Wuhan) Wuhan, China
B.Eng. in Intelligent Science and Technology; Advisor: ***Prof. Chang Tang***
Sep. 2020 – Jun. 2024 *Average Score (100-point scale): 86.97/100*

PUBLICATIONS

1. **Zixiao Jin**, Xiao Zheng, Weiqing Yan, Chang Tang. CIPR: Class Imbalance-Aware Pseudo-Label Regeneration for Single-Cell Multi-Omics Clustering. **IEEE Transactions on Knowledge and Data Engineering**. Early Access, pp. 1–11, 2026, DOI: 10.1109/TKDE.2026.3695648. (**CCF A**, JCR Q1)
2. **Zixiao Jin**, Xiao Zheng, Chang Tang, Chuankun Li, Yuanyuan Liu, Xinwang Liu. Task-Aware Information Decoupling for Multimodal Clustering. **IEEE Transactions on Knowledge and Data Engineering**. vol. 38, no. 6, pp. 3961–3974, June 2026, DOI: 10.1109/TKDE.2026.3679087. (**CCF A**, JCR Q1)
3. **Zixiao Jin**, Xiao Zheng, Hua Zhou, Chengfu Ji, Sen Xiang, Chang Tang. DD-HGNN+: Drug-Disease Association Prediction via General Hypergraph Neural Network With Hierarchical Contrastive Learning and Cross Attention Learning. **IEEE Journal of Biomedical and Health Informatics**. vol. 29, no. 11, pp. 7810–7819, November 2025, DOI: 10.1109/JBHI.2025.3542784. (**CCF C**, JCR Q1)
4. **Zixiao Jin**, Minhui Wang, Xiao Zheng, Jiajia Chen, Chang Tang. Drug side effects prediction via cross attention learning and feature aggregation. **Expert Systems with Applications**. vol. 248, Art. no. 123346, August 2024, DOI: 10.1016/j.eswa.2024.123346. (**CCF C**, JCR Q1)
5. **Zixiao Jin**, Minhui Wang, Chang Tang, Xiao Zheng, Wen Zhang, Xiaofeng Sha, Shan An. Predicting miRNA-disease association via graph attention learning and multiplex adaptive modality fusion. **Computers in Biology and Medicine**. vol. 169, Art. no. 107904, February 2024, DOI: 10.1016/j.compbiomed.2023.107904.
6. **Zixiao Jin**, Dong Yang, Zhe Liu, Xiao Zheng, Jiahe Jiang, Chang Tang. Decoupled multimodal fusion for miRNA–disease association identification. **Biomedical Signal Processing and Control**, vol. 127, Art. no. 111167, 2026 (in press). DOI: 10.1016/j.bspc.2026.111167. (JCR Q2)
7. Xiao Zheng, **Zixiao Jin**, Zhiwei Ye, Chang Tang, Xinwang Liu. Robust Sample Relationship-Guided Contrastive Multi-view Clustering. **IEEE Transactions on Multimedia**. Accepted, 2026. (**CCF A**, JCR Q1)

MANUSCRIPTS UNDER REVIEW

1. Xiao Zheng, Wei Feng, Tong Yang, **Zixiao Jin**, Zhiwei Ye, Chang Tang, Xinwang Liu. Dual-Graph Collaborative Clustering Network for Unsupervised Hyperspectral Band Selection. **IEEE Transactions on Circuits and Systems for Video Technology**. (**CCF B**, JCR Q1)
2. Shuhan Huang, **Zixiao Jin**, Jiahe Jiang, Chang Tang, Xiao Zheng, Yang Li. RHGAT: A relational information fused heterogeneous graph attention network for drug-target interaction prediction. **IEEE Transactions on Computational Biology and Bioinformatics**. (**CCF B**, JCR Q1)
3. Changming Yao, **Zixiao Jin**, Chang Tang, Xinzhong Zhu. Mitigating Representation Collapse in Drug Side Effect Frequency Prediction. **Neural Networks**. (**CCF B**, JCR Q1)

INTELLECTUAL PROPERTY

- **Inventor**. Drug Side-Effect Prediction Method, Apparatus, Electronic Device, and Storage Medium. **Granted Chinese Invention Patent**, Application No. 202310702073.4, filed Jun 14, 2023.
- **Co-copyright Holder**. Drug Side-Effect Prediction System (ADR Prediction System), V1.0. Computer Software Copyright Registration No. 2023SR0984726, registered Aug 30, 2023.

SELECTED RESEARCH EXPERIENCE

Task-Aware Information Decoupling for Multimodal Clustering 2025 – Present

***First Author** | **IEEE TKDE** (2026) | NSFC-supported*

- Led the development and empirical evaluation of DRLMMC to address a limitation of task-agnostic cross-modal alignment: it can retain redundant modality noise and obscure task-relevant information.
- Designed modality-specific encoders with mutual-information constraints and combined them with contrastive cross-modal alignment and conditional mutual-information projection to disentangle shared representations from task-relevant modality-specific representations.
- Evaluated DRLMMC on 9 multimodal benchmark datasets and 2 single-cell multi-omics datasets; on CCV, it achieved a clustering accuracy (ACC) of **54.02%**, outperforming the best-performing baseline (MFLVC) by **22.79 pp**.

Zero-Shot Skeleton-Based Action Recognition May 2026 – Present

*Research Intern, EIT-iSPH Labs, supervised by **Prof. Jie Liang**, Eastern Institute of Technology, Ningbo (EIT)*

- Developed a conflict-aware framework that preserves shared and description-specific information across motion-focused and global textual descriptions and combines feature- and decision-level evidence through a residual adapter with only 290 trainable parameters.
- Preliminary results averaged over 5 random seeds outperformed the reproduced Flora baseline across all 4 evaluated zero-shot protocols on the NTU RGB+D 60 and 120 benchmarks; the largest gain occurred on the NTU RGB+D 120 96/24 split (**68.873%** vs. 64.836%; **+4.037 pp**).

Class-Imbalance-Aware Single-Cell Multi-Omics Clustering 2025 – Present

***First Author** | **IEEE TKDE** (Early Access, 2026) | NSFC-supported*

- Developed an optimal-transport formulation for pseudo-label regeneration under class imbalance to mitigate majority-class dominance during cross-omics alignment.
- Introduced a soft-uniform prior and curriculum learning to limit noisy pseudo-label propagation, and designed a cross-omics prediction-interaction module to reduce modality conflicts.
- For tail classes in PBMC-10k, achieved **77.41% Macro-F1** and **72.87% tail-class accuracy (Tail-ACC)**, outperforming the best-performing baseline (scMDCL) by **7.69** and **4.92** percentage points, respectively.

Multi-View Clustering for Spatial Transcriptomics Jul 2025 – Present

***Project Lead** | University Innovation Training Program (General Project), Project No. 2025XLB83 | Midterm Evaluation Passed*

- Lead a 6-member student team to develop a multi-view clustering framework integrating gene expression, spatial coordinates, and histology images for spatial domain identification.
- Reproduced the OmiCLIP visual–omics training and fine-tuning pipeline used in the Loki platform (*Nature Methods*, 2025) on paired H&E images and spatial transcriptomic profiles, establishing a reproducible baseline for the project.
- Adapted a CLIP-PGS-inspired patch-masking strategy (CVPR 2025) for OmiCLIP fine-tuning to test whether suppressing redundant tissue patches while retaining morphology-relevant evidence improves image–transcriptome alignment; initial masked-versus-unmasked experiments are complete, with broader evaluation ongoing.

Graph-Based Biomedical Association Prediction 2023 – 2025

***First Author** on 4 Journal Papers arising from this work (2024–2026)*

- Developed graph-based and multimodal representation-learning models for predicting drug–disease associations, drug side-effect frequencies, and miRNA–disease associations.
- Combined hypergraph neural networks, hierarchical contrastive learning, cross-attention mechanisms, sparsity-aware loss functions, and adaptive modality fusion to model sparse, higher-order biomedical associations.
- Published 3 first-author papers in IEEE JBHI, Expert Systems with Applications, and Computers in Biology and Medicine, with a fourth in press at Biomedical Signal Processing and Control.

SELECTED AWARDS AND HONORS

- **First-Class Graduate Academic Scholarship**, China University of Geosciences (Wuhan), 2024 and 2025.
- **Outstanding Graduate and Outstanding Graduation Thesis**, China University of Geosciences (Wuhan), 2024.
- **Huawei Scholarship and President Scholarship**, China University of Geosciences (Wuhan), 2023.
- **National Second Prize**, 13th MatherCup University Mathematical Modeling Competition, 2023.
- **National Second Prize**, 11th Teddy Cup Data Mining Challenge, 2023.
- **National First Prize and National Second Prize**, Advanced Robot Vision, China Robot Competition and RoboCup China Open, 2022.
- **Provincial First Prize / National Second Prize Nomination**, Contemporary Undergraduate Mathematical Contest in Modeling, 2022.

TECHNICAL SKILLS

Programming languages: Python, C/C++, Java.
Machine learning and research tools: PyTorch, Git, LaTeX; C++/Qt for cross-platform application development.
Languages: Mandarin Chinese (native); English (CET-6: 448; academic reading and writing).